

BASHAR IBRAHEM

Autonomous AI & Machine Learning Engineer | Agentic Systems Architect

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PROFESSIONAL SUMMARY

Autonomous AI & Machine Learning Engineer specializing in multi-agent DAG orchestration, enterprise LLM safety guardrails, and production ML pipelines. Winner/Finalist in 4 premier AI hackathons (Google All Things Agentic, Experian Digital Transformation, Microsoft RektHackathon, and AI Tinkerers KL). Proven record architecting MCP-native agent systems, hardware TEE execution guardrails, self-healing prompt optimization engines, and containerized microservices on Google Cloud Run, FastAPI, and Docker.

TECHNICAL SKILLS & CORE COMPETENCIES

Agentic AI & LLMs:	Google Gemini 3.7/2.5 Flash, Model Context Protocol (MCP), Multi-Agent DAG State Machines, Autonomous Tool/Function Calling, Constitutional AI Guards, Vector RAG (FAISS), Evolutionary Prompt Hardening.
Agent Safety & TEE:	Hardware TEE Enclaves (Terminal 3 / Intel SGX), DID & Verifiable Credentials (VCs), OWASP LLM Top 10 Red-Teaming, Prompt Injection Hardening, Client-Side Zero-PII Scrubbing (100% PDPA Compliant).
Machine Learning & MLOps:	Supervised & Unsupervised Learning, Random Forest, XGBoost, Scikit-Learn, Deep Learning (TensorFlow/Keras), Time-Series (ARIMA), MLflow Experiment Tracking, Pandera Data Validation.
Backend, Cloud & CI/CD:	Python (FastAPI, Uvicorn, Typer, uv), Node.js (pnpm), Google Cloud Run, Cloud Firestore, Cloud Storage, Docker, GitHub Actions CI/CD, Server-Sent Events (SSE), SQLite, Supabase Auth.
Frontend & UI/UX:	React 18, Next.js 14, Vite, TypeScript, Tailwind CSS, Recharts, Web Speech API (Voice Synthesis).
Testing & Tooling:	Pytest (100% Pass Rate), Antigravity/Cursor Toolchain Automation, Postman, Linux/Unix, Vercel Production.

CHAMPIONSHIP AI SYSTEMS & KEY ENGINEERING PROJECTS

AutoGuard AI — Autonomous AI Red-Teaming & Prompt Hardening Taskmaster [\[Live\]](#) [Google Hackathon Winner | 2026](#) [Demo](#) | [GitHub](#) | [Marketplace Action](#)

Python • Google Gemini 3.7 Flash • FastAPI • Google Cloud Run • MCP Protocol • GitHub Actions • Pytest

- **Autonomous 6-Stage DAG Execution:** Architected a multi-agent SecOps Taskmaster using Gemini 3.7 Flash and FastAPI that automates adversarial red-teaming and prompt patching, shrinking manual 3-week penetration audits to **<30 seconds**.
- **Evolutionary Model Armor:** Engineered an optimization engine that synthesizes 50+ adversarial attack vectors mapped to OWASP LLM Top 10, boosting system prompt safety resilience from **28.4% to 98.6% (Grade A+)**.
- **MCP Server & Marketplace Delivery:** Published an official Model Context Protocol (MCP) server for Antigravity/Cursor/Claude and a verified GitHub Marketplace Action ([v1.0.0](#)) backed by **21/21 (100%) automated test coverage** on Cloud Run.

VaultPay — Autonomous B2B Procurement Agent with Hardware TEE Enclave [AI Tinkerers KL × Terminal 3 Finalist | 2026](#) [Guardrails](#) [\[GitHub\]](#)

TypeScript • Terminal 3 ADK • Intel SGX TEE • Verifiable Credentials (DID) • Next.js • Node.js

- **Hardware-Enforced Guardrails:** Architected an autonomous agent with zero access to payment secrets, delegating financial execution to an isolated Trusted Execution Environment (TEE) enforcing strict \$1,000 per-call caps and vendor allowlists.
- **Signed Agent-to-Agent Handoff:** Implemented bilateral negotiation between Buyer and Supplier agents, exchanging cryptographically signed quote receipts (#QUOTE-...) and nonces before any money moves.
- **Verifiable KYC & Tamper-Evident Ledger:** Integrated selective-disclosure DID proofs for buyer role authorization and a SHA-256 chained audit ledger guaranteeing 100% immutable transaction verifiability.

SmartFlow One — AI Financial Document Intelligence & PDPA Protection [\[Live Demo\]](#) | [Experian Hackathon Champion](#) | 2026 [GitHub](#)

Next.js · React · Google Gemini 2.5 Flash · FAISS Vector Index · Web Speech API · Supabase

- **Zero-PII Client-Side Masking:** Developed a deterministic privacy engine that scrubs NRICs, credit cards, and confidential variables before cloud dispatch, achieving **100% PDPA compliance**.
- **Vector Semantic Extraction:** Built an end-to-end processing pipeline utilizing FAISS vector embeddings and Gemini 2.5 Flash to extract revenue, OpEx, and net profits with page-level citation audit trails in **<15 seconds**.
- **Audio Executive Briefing:** Integrated the Web Speech API to condense dense financial disclosures into a 75-word synthesized voice briefing narrated aloud in ~30 seconds.

RakyatOS — Grounded Civic AI & Anti-Scam Intelligence OS [\[Live Demo\]](#) | [GitHub](#) | [Microsoft RektHackathon Finalist](#) | 2026

Python · FastAPI · Qwen LLM · PDRM Semak MuLe API · React · Tailwind CSS

- **Proactive ScamShield Screening:** Engineered a pre-attack threat filter cross-referencing incoming messages against PDRM Semak Mule and NSRC 997 fraud registries to combat RM2.97B online scam losses.
- **Anti-Hallucination Hard-Gate:** Implemented a deterministic verification barrier anchoring answers to official government portals (MyGOV, STR/LHDN, SSM), enforcing **0% unverified citations**.

RainToday — Production Malaysian Hourly Rainfall Forecasting System [\[Live Demo\]](#) | [Production ML System](#) | 2025 – 2026 [GitHub](#)

Python · Scikit-Learn · FastAPI · SQLite · Pandera · MLflow · React · Vite

- **End-to-End ML Pipeline:** Trained Random Forest classification and regression models across 7 Malaysian meteorological stations, achieving **83.15% accuracy, 77.23% F1-score, and 88.33% ROC-AUC**.
- **Production Serving & Quality Gate:** Deployed a modular FastAPI backend with a 60-minute SQLite caching layer to prevent API rate-limits and Pandera validation enforcing climate bounds.

VERIFIED PROFESSIONAL CERTIFICATIONS

IBM — Machine Learning with Python

Supervised/Unsupervised Learning, Regression, Clustering, Scikit-Learn ([Verify: WB4CTI3FHT5P](#))

KodeKloud — ML Model Development & Tracking (MLOps)

Experiment Logging, Model Versioning, Artifact Registry, MLflow ([Verify: S6GPUQ3LSXE5](#))

IBM — Databases & SQL for Data Science with Python

Relational DB Queries, Multi-table Joins, Python DB-API ([Verify: LH6Z63ORZ6CL](#))

IBM — Deep Learning & Neural Networks with Keras

Multi-Layer Perceptrons, Backpropagation, TensorFlow/Keras ([Verify: 22M00JV3SZ98](#))

KodeKloud — Data Engineering Essentials

ETL Pipelines, Schema Validation, Pandera Data Quality ([Verify: HQC516DPB9JK](#))

EDUCATION & HONORS

Bachelor of Science in Computer Science / AI Focus

Malaysia

- **Core Coursework:** Machine Learning, Deep Neural Networks, Distributed Systems, Algorithms & Data Structures, Relational Database Systems.
- **Championship Honors:** Winner / Finalist across Google All Things Agentic Hackathon, Experian Digital Transformation Hackathon, Microsoft RektHackathon, and AI Tinkerers KL 2026.